

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

```
samalutina@samalutina:~$ ls /etc/ > file.txt
samalutina@samalutina:~$ ls >> file.txt
samalutina@samalutina:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
```

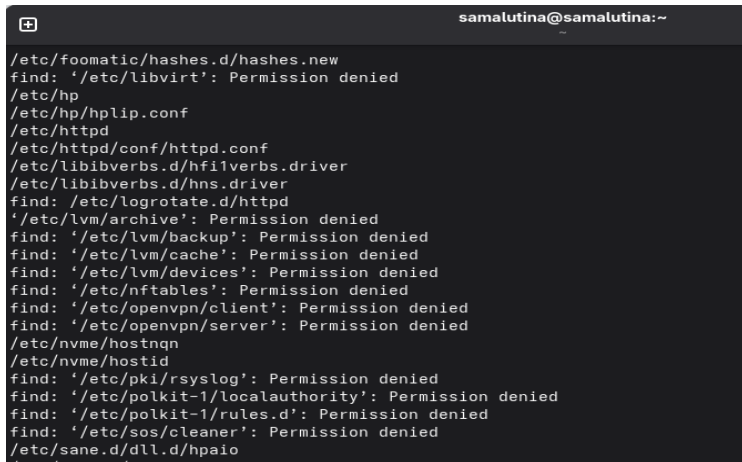
Рисунок 1: Запись в файл

```
samalutina@samalutina:~$ grep .conf file.txt > conf.txt
samalutina@samalutina:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
```

Рисунок 2: Поиск расширения .conf

```
samalutina@samalutina:~$  
samalutina@samalutina:~$ ls -R | grep c*  
conf.txt  
samalutina@samalutina:~$ find ~ -name "c*" -print  
/home/samalutina/.cache/evolution/calendar  
/home/samalutina/.cache/gnome-software/appstream/components.xmlb  
/home/samalutina/.cache/gnome-software/flatpak-system-default/components.xmlb  
/home/samalutina/.cache/gnome-software/flatpak-user-user/components.xmlb  
/home/samalutina/.cache/gnome-software/icons/c058f4d673e8f74a1f4a3e74cf27b8139ee4021a-re.sonny.Tangram.png  
/home/samalutina/.cache/gnome-software/icons/cceac35c311511cc9da74bd0162bcf8f0fb0f27c-org.gaphor.Gaphor.png  
/home/samalutina/.cache/gnome-software/icons/c836206838ff53df7f9498c14fff5c510ff89ee5-no.bragefuglseth.Keypunch  
.png  
/home/samalutina/.cache/gnome-software/icons/c188551f4fa6e0e32072a589ef729f820f2c2aff-net.nokyan.Resources.png  
/home/samalutina/.cache/gnome-software/icons/c105f1c77c90ca8b6a3e76caed60a87265008e47-io.github.revisto.drum-ma  
chine.png  
/home/samalutina/.cache/gnome-software/icons/cfa29742ffffa7ac312607c78c63722ec17f96e2-de.haeckerfelix.Shortwave  
.png  
/home/samalutina/.cache/gnome-software/icons/c85fd92d7a71b89002526eee10c44517427e2eb5-com.github.huluti.Curtail  
.png  
/home/samalutina/.cache/gnome-software/icons/ce763f7a3c7686b3f884adb3cfd056ff3a03768-org.gnome.DejaDup.png  
/home/samalutina/.cache/gnome-software/icons/c145154d07311aa67aad50a45675ec3f8fed5b02-page.tesk.Refine.png
```

Рисунок 3: Поиск файлов



```
samalutina@samalutina:~  
/etc/foomatic/hashe.s.d/hashe.s.new  
find: '/etc/libvirt': Permission denied  
/etc/hp  
/etc/hp/hplip.conf  
/etc/httpd  
/etc/httpd/conf/httpd.conf  
/etc/libibverbs.d/hfi1verbs.driver  
/etc/libibverbs.d/hns.driver  
find: /etc/logrotate.d/httpd  
'/etc/lvm/archive': Permission denied  
find: '/etc/lvm/backup': Permission denied  
find: '/etc/lvm/cache': Permission denied  
find: '/etc/lvm/devices': Permission denied  
find: '/etc/nftables': Permission denied  
find: '/etc/openvpn/client': Permission denied  
find: '/etc/openvpn/server': Permission denied  
/etc/nvme/hostnqn  
/etc/nvme/hostid  
find: '/etc/pki/rsyslog': Permission denied  
find: '/etc/polkit-1/localauthority': Permission denied  
find: '/etc/polkit-1/rules.d': Permission denied  
find: '/etc/sos/cleaner': Permission denied  
/etc/sane.d/dll.d/hpaio
```

Рисунок 4: Поиск файлов

```
samalutina@samalutina:~$  
samalutina@samalutina:~$ find /etc -name "h*" -print | less  
samalutina@samalutina:~$  
samalutina@samalutina:~$  
samalutina@samalutina:~$ find ~ -name "log*" > logfile &  
[1] 20068  
samalutina@samalutina:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
samalutina@samalutina:~$ rm logfile  
samalutina@samalutina:~$
```

Рисунок 5: Фоновый запуск процесса

```
samalutina@samalutina:~$  
samalutina@samalutina:~$ gedit &  
[1] 20117  
samalutina@samalutina:~$ ps | grep gedit  
20117 pts/0    00:00:00 gedit  
samalutina@samalutina:~$ kill 20117  
[1]+  Terminated                  gedit  
samalutina@samalutina:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

```
DE(1)                                User Commands                                DE(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df.  df displays the amount of space available on the
    file system containing each file name argument.  If no file name is given, the space available on all
    currently mounted file systems is shown.  Space is shown in 1K blocks by default, unless the environ-
    ment variable POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

    If an argument is the absolute file name of a device node containing a mounted file system, df shows
    the space available on that file system rather than on the file system containing the device node.
    This version of df cannot show the space available on unmounted file systems, because on most kinds of
    systems doing so requires non-portable intimate knowledge of file system structures.

OPTIONS
    Show information about the file system on which each FILE resides, or all file systems by default.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        include pseudo, duplicate, inaccessible file systems

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes;
        see SIZE format below

    --direct
    Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

```
du(1)                                     User Commands                               du(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILES, recursively for directories.

    Mandatory arguments to long options are mandatory for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size is usually smaller, it
        may be larger due to holes in ('sparse') files, internal fragmentation, indirect blocks, and the
        like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes;
        see SIZE format below

    -b, --bytes
        equivalent to '--apparent-size --block-size=1'

Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
samalutina@samalutina:~$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 37142260 110878540 26% /
devtmpfs         4011292      0    4011292  0% /dev
tmpfs            4035664      92    4035572  1% /dev/shm
tmpfs           1614268     1952    1612316  1% /run
tmpfs             1024        0        1024  0% /run/credentials/systemd-journald.service
tmpfs           4035668    136800    3898868  4% /tmp
/dev/nvme0n1p3 155186176 37142260 110878540 26% /home
/dev/nvme0n1p2 1992552     518800    1352512 28% /boot
/dev/loop1       115712     115712        0 100% /var/lib/snapd/snap/hugo/25676
/dev/loop0        68480      68480        0 100% /var/lib/snapd/snap/core24/1349
/dev/loop2        49280      49280        0 100% /var/lib/snapd/snap/snapd/25935
tmpfs             1024        0        1024  0% /run/credentials/systemd-resolved.service
tmpfs           807132      188     806944  1% /run/user/1088
samalutina@samalutina:~$
```

Рисунок 9: Справка по команде du

```
4      ./git-extended/.git/objects/e2
4      ./git-extended/.git/objects/76
4      ./git-extended/.git/objects/55
4      ./git-extended/.git/objects/db
4      ./git-extended/.git/objects/e0
4      ./git-extended/.git/objects/13
4      ./git-extended/.git/objects/2c
76     ./git-extended/.git/objects
8      ./git-extended/.git/logs/refs/heads
8      ./git-extended/.git/logs/refs/remotes/origin
8      ./git-extended/.git/logs/refs/remotes
16     ./git-extended/.git/logs/refs
20     ./git-extended/.git/logs
212    ./git-extended/.git
220    ./git-extended
0      ./monthly
0      ./reports/monthly/monthly
0      ./reports/monthly
0      ./reports
4      ./ski.plases/equipment
0      ./ski.plases/plans
4      ./ski.plases
0      ./australia
0      ./play/games/play
0      ./play/games
0      ./play
788424 .
samalutina@samalutina:~$
```

Рисунок 10: Запуск команды du

```
/home/samalutina/git-extended/.git/objects/6c
/home/samalutina/git-extended/.git/objects/2d
/home/samalutina/git-extended/.git/objects/ba
/home/samalutina/git-extended/.git/objects/0e
/home/samalutina/git-extended/.git/objects/e2
/home/samalutina/git-extended/.git/objects/76
/home/samalutina/git-extended/.git/objects/55
/home/samalutina/git-extended/.git/objects/db
/home/samalutina/git-extended/.git/objects/e0
/home/samalutina/git-extended/.git/objects/13
/home/samalutina/git-extended/.git/objects/2c
/home/samalutina/git-extended/.git/logs
/home/samalutina/git-extended/.git/logs/refs
/home/samalutina/git-extended/.git/logs/refs/heads
/home/samalutina/git-extended/.git/logs/refs/remotes
/home/samalutina/git-extended/.git/logs/refs/remotes/origin
/home/samalutina/monthly
/home/samalutina/reports
/home/samalutina/reports/monthly
/home/samalutina/reports/monthly/monthly
/home/samalutina/ski.plases
/home/samalutina/ski.plases/equipment
/home/samalutina/ski.plases/plans
/home/samalutina/australia
/home/samalutina/play
/home/samalutina/play/games
/home/samalutina/play/games/play
samalutina@samalutina:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.